



Department of Information Technology

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CD3291 & Data Structures and Algorithms

Name of the Faculty member (s): Mrs. G. Mareeswari, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit V / Topological Ordering
- **Course Outcome:** CO5
- **Activity Chosen:** Think-Pair- Share
- **Justification:**

Topological ordering was chosen for the Think-Pair-Share activity to encourage active learning and collaborative problem-solving in understanding graph-based concepts. The topic involves arranging vertices of a directed acyclic graph (DAG) in a linear order such that every directed edge from vertex u to vertex v places u before v . This concept can be challenging for students to visualize and apply individually. Working in pairs enables students to discuss the logic behind dependencies, clarify misconceptions, and verify the correctness of their orderings. Through peer interaction, they reinforce theoretical knowledge and develop algorithmic reasoning skills. Topological ordering has practical relevance in areas like task scheduling, course prerequisite planning, and compiler design, making it both engaging and applicable.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - ✓ The instructor begins by explaining the concept of **topological ordering**, highlighting its importance in directed acyclic graphs (DAGs). Students are advised to review related materials and algorithms in prior.
 - ✓ **Think (5 Minutes):** Each student individually thinks and attempt to outline the possible topological order for the given example.
 - ✓ **Pair (5 Minutes):** Students pair up to discuss and compare their graphs and orderings. They collaborate to identify errors such as cycles or incorrect dependencies, refine their understanding of in-degree calculation

- ✓ **Share (Class Discussion):** Each pair presents their DAG and the derived topological order to the class.
- ✓ Finally share points with the entire class (Share) as shown in Fig 1 & 2
- ✓ **Outcome:** Students gain practical understanding of graph dependencies, enhance algorithmic reasoning, and develop collaboration and problem-solving skills through active learning.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO5	3	1	1	1	2	1

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	1	1	1	2	1
Justification for correlation	Requires basic engineering knowledge for understand DAG concept	Different algorithms applied for analyzing and solving complex engineering problems using DAG	Able to design solutions for the complex nonlinear problems using the concept of DAG.	Able to solve problem in team in this activity.	Able to explain the way to solve the problem using DAG with their team members.	The students can apply the concept of DAG in solving various problems of data science and algorithms.

• **Images / Screenshot of the practice:**



Figure1a: Students discussion with their team members [DanyaSankari M,A. Varuna Shree, S Santhiya Iswarya, K Kavya]

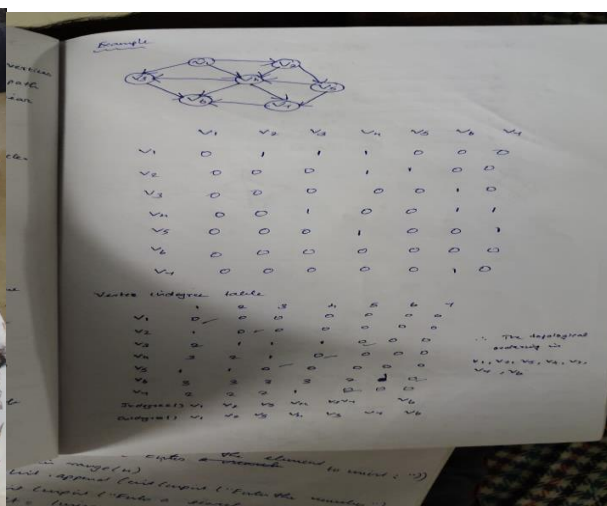


Figure1b: Solutions shared by team members [DanyaSankari M,A. Varuna Shree, S Santhiya Iswarya, K Kavya]

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- All students respond that they actively involved in the activity and discussions.
 - Later, students mentioned that they explored and exchanged ideas about topological ordering with their peers.
 - Students also informed that pairing up allowed them to learn from each other and clarify their doubts. They also shared that working in pairs helped them learn collaboratively and clear their doubts effectively.

- ❖ ***Benefit of the practice:***

- The activity increased students' confidence in understanding and explaining the concept of Directed Acyclic Graphs.
 - It also helped them clearly understand how to use Topological Ordering in real-life applications.

- ❖ ***Challenges faced in implementation:***

- Few students participated less actively in the discussions, which affected the overall group interaction.
 - Some students found it difficult to express their ideas clearly, leading to misunderstandings within their teams.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/3_GM_CS3353_Evaluation_Expression_TPS.pdf
 - ❖ <https://www.theteachertoolkit.com/index.php/tool/think-pair-share>
https://www.ritrjpm.ac.in/images/computer-science/2021-2022/MGM_VI_Think_Pair_and_Share.pdf

Signature of Faculty Member

HOD